

Property Walay

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Chapter 1

Introduction

PropertyWalay is an AI-powered real estate chatbot we've built specifically for Pakistan's property market. We wrote this document to help everyone involved understand what we're building and why. If you're a developer, you'll find all the technical details about how our chatbot and AI recommendation engine work. Project managers can use this to see how we're doing and make sure we're on track. Our marketing team will learn about what makes our product special and who we're building it for. Property seekers and agents will understand what the platform can actually do for them. And testers can use this to figure out what needs testing.

1.1 Problem Statement

Aspect	Description
Problem	Let's be honest - finding property in Pakistan is a real headache. You're juggling multiple websites like Zameen.com, Graana, OLX, and even random Facebook groups, hoping to find something decent. The worst part? Half the listings you see are already sold or rented out, but they're still sitting there online. And if you're more comfortable in Urdu or Roman Urdu, tough luck - most platforms are English-only.
Affects	Think about it from both sides. Property seekers are wasting hours, sometimes days, calling agents about properties that aren't even available anymore. Meanwhile, agents are pulling their hair out answering the same basic questions over and over - "What's the price?" "Is it still available?" - instead of actually working with serious buyers. Nobody has the real-time information they need.

Aspect	Description
Impact	This mess is actually holding back our entire real estate sector. People spend weeks trying to find a simple flat or house. Agents are losing money because they're stuck dealing with endless casual inquiries. And a huge chunk of the population just can't use these platforms comfortably because of the language barrier.
A Successful Solution Would Be	We need something that brings it all together - one smart platform that pulls in live listings from all the major sites and actually understands what you're asking for in plain language. Whether you speak English or prefer Roman Urdu, even if you just want to use voice search, it should work for you. The system should keep track of properties you like and tell you when prices change or if they get sold. And it should make it dead simple to schedule meetings with agents without all the back-and-forth.

Table 1.1: Detailing the Problem Statement

1.2 Scope

PropertyWalay is here to shake things up in Pakistan's property market. We're fixing the real problems people face every day. The platform connects directly to Zameen.com, Graana.com, and Booking.com to give you access to everything that's actually available right now. Our AI chatbot gets what you're saying whether you type in English or Roman Urdu - no more struggling with complicated search filters. The chatbot uses some pretty cool retrieval-augmented generation technology to learn what you like and suggest properties that actually match what you're looking for. You can save properties you're interested in, and we'll automatically ping you if the price drops, if there's an update, or if it gets sold. We're also analyzing what's hot in the market so you can see which areas are trending - super useful if you're thinking about investing. For people living abroad or anyone who can't physically visit properties, we generate video tours that show you around and give you context about the location. We've automated the whole meeting scheduling thing between buyers and agents, so no more phone tag. Agents can generate professional PowerPoint presentations instantly for their client meetings. The system remembers everything you've searched for, so you can pick up right where you left off. Voice search works if you prefer talking over typing. You can use it on your computer or phone - whatever works for you. We take your privacy seriously and use proper encryption to keep your data safe. And we've built this to scale as Pakistan's real estate market keeps growing.

1.3 Modules

1.3.1 Natural Language Understanding (NLU) Module

This is the brain that lets you talk to the chatbot like a normal person. You can describe what you're looking for in everyday language, whether that's English or Roman Urdu. No need to mess around with those annoying filter dropdowns.

1. Process natural language queries in English and Roman Urdu
2. Extract key search parameters from conversational text
3. Support voice-to-text conversion for hands-free searching

1.3.2 AI-Powered Recommendation Engine

This module is what makes our recommendations actually useful. It uses retrieval-augmented generation and machine learning to understand what you really want. The more you use it, the smarter it gets at finding properties you'll actually like.

1. Generate personalized recommendations using RAG technology
2. Rank results based on user preferences and behavior
3. Refine recommendations based on user feedback

1.3.3 Real-Time Listings Integration Module

This solves that annoying problem where you call about a property and find out it was sold two weeks ago. We're constantly pulling fresh data from major Pakistani real estate websites through APIs and web scraping. What you see is what's actually available.

1. Fetch and synchronize listings from Zameen.com, Graana.com, and Booking.com
2. Update property data in real-time for availability and price changes
3. Standardize data formats across different source platforms

1.3.4 Property Tracking and Notification Module

Save properties you're interested in and let us watch them for you. We'll notify you the moment something changes - price drops, new updates, or if it gets sold. No more checking the same listings ten times a day.

1. Allow users to save and bookmark properties of interest
2. Monitor saved properties for price changes
3. Send notifications when properties are marked as sold or unavailable

1.3.5 Investment Insights and Trends Module

If you're thinking about investing, this module tells you what's hot and what's not. We analyze all the market data to show you which areas are getting attention and where prices

are moving. It's like having your own market research team.

1. Analyze listing activity patterns to identify trending neighborhoods
2. Generate heat maps showing property demand across different areas
3. Compare price trends across locations and property types

1.3.6 Meeting Automation Module

Tired of playing phone tag with agents? This module handles all the scheduling for you. Just tell the chatbot when you want to meet, and we'll coordinate with the agent, find a time that works, and send reminders to both of you.

1. Enable users to request meetings with agents through the chatbot
2. Integrate with agent calendars to find available time slots
3. Send automated meeting confirmations and reminders to both parties

1.3.7 Automated Presentation Generation Module

For agents, this is a game-changer. Instead of spending hours putting together PowerPoint presentations for clients, you can generate professional-looking presentations in seconds. More time for actual selling, less time on busywork.

1. Generate PowerPoint presentations with property details on demand
2. Include property images, floor plans, and location maps
3. Format presentations with professional templates

1.3.8 Location-Based Video Generation Module

Can't visit a property in person? We create video tours that show you what the property looks like, where it's located, and what's nearby. This is especially helpful if you're overseas or just can't make the trip.

1. Generate short video tours using property images and data
2. Overlay location information and nearby amenities
3. Include map visualizations showing property location

1.3.9 User History and Profile Management Module

We remember everything - what you've searched for, which properties you've looked at, what you seem to prefer. This means the platform gets better at helping you the more you use it. You can also go back and check out properties you looked at last week or last month.

1. Store user search queries and browsing history across sessions
2. Maintain lists of viewed, saved, and contacted properties
3. Build user preference profiles based on interaction patterns

1.3.10 Multilingual and Voice Interface Module

Everyone should be able to use this platform comfortably. Switch between English and Roman Urdu whenever you want. Use voice search if you're driving or just prefer talking. We even handle it when you mix English and Urdu words in the same sentence - because that's how people actually talk.

1. Support seamless switching between English and Roman Urdu
2. Implement voice recognition for hands-free property searching
3. Handle code-mixing where users combine English and Urdu words

1.4 User Classes and Characteristics

User class	Description
Property Seekers	These are everyday people like you and me looking for a place to live or buy. They're about 80% of who uses the platform - could be a young professional hunting for an apartment, or a family trying to buy their first house. A lot of them are more comfortable with Roman Urdu. They're usually searching pretty actively, maybe 5-10 times a week. The common thread? They're short on time, frustrated with seeing properties that are already gone, and just want straight, honest, up-to-date information.
Real Estate Agents	These are the licensed dealers who list properties and help close deals. We're talking about 2,000-3,000 active agents across Pakistan's major cities, and they're working long days - easily 6-8 hours managing listings and answering calls. Their biggest headache? Endless calls from people who are just casually browsing, not serious buyers. What they really need are features that help them spot who's actually ready to buy, schedule meetings without all the back-and-forth, and whip up professional presentations quickly.
Investors and Developers	These folks are looking at properties as business opportunities, not homes. They make up about 15% of our users. Usually pretty well-educated with solid financial backgrounds, they want hard data on market trends and up-and-coming neighborhoods. They're not in a rush - checking in maybe 2-3 times a week to do their research. They care way more about ROI and investment potential than whether the kitchen is nice. They expect serious analytics and professional market reports.

User class	Description
Overseas Pakistanis	Pakistanis living abroad who want to buy or rent back home - they're about 20% of property seekers. The big challenge for them is being thousands of miles away and not being able to just go see a property. They really depend on video tours, detailed descriptions, and anything that gives them a virtual view. Trust is huge for them since they can't easily verify things in person. They're usually logging in during their evening hours, which might be our nighttime.

Table 1.2: User Classes and Characteristics

Chapter 2

Project Requirements

In the rapidly evolving landscape of Pakistan’s real estate market, PropertyWalay emerges as a transformative AI-powered chatbot system designed to revolutionize property searching through intelligent automation and multilingual support. This chapter introduces the foundational elements that guide the structure and functionality of PropertyWalay in a detailed yet comprehensive manner. The use of Use Case Diagrams, Event Response Tables, and clear definitions of Functional Requirements (FRs) and Non-Functional Requirements (NFRs) ensures that all stakeholders—from property seekers and real estate agents to developers and platform administrators—have a thorough understanding of the system’s capabilities and expectations.

The intelligent design of PropertyWalay allows for a meticulous integration of artificial intelligence, natural language understanding, and real-time data aggregation into a seamless property search experience. By addressing the fragmented nature of existing platforms and bridging the communication gap through Roman Urdu support, PropertyWalay transforms how Pakistanis discover and interact with property listings. Use Case Diagrams provide a visual representation of the system’s interactions with different stakeholders, facilitating an intuitive understanding of various functionalities and user scenarios. Event Response Tables outline the specific actions the system will undertake in response to different events, ensuring predictable, efficient, and user-centric operation.

Together, these elements form a robust framework that supports the detailed documentation and specification needed to develop a system that not only meets but exceeds the demands of Pakistan’s modern real estate sector. PropertyWalay’s requirements are structured to deliver personalized recommendations, automated tracking, investment insights, and seamless agent coordination—all within a conversational interface that feels natural to Pakistani users.

2.1 Use-case/Event Response Table

For this project, we utilized a combination of a use case diagram and an event response table as our primary requirement gathering techniques. This section elaborates on the methods selected to comprehensively capture and define PropertyWalay’s requirements, ensuring a clear understanding of user interactions and system responses. These techniques facilitated a structured approach to identifying and documenting the necessary functionalities and behaviors of the AI-powered real estate chatbot system.

The use case diagram provides a concise overview of PropertyWalay’s capabilities, helping stakeholders understand its core offerings and user interactions by visually capturing the platform’s primary interactions with pertinent roles associated with the system—including Property Seekers, Real Estate Agents, Platform Administrators, and external service providers.

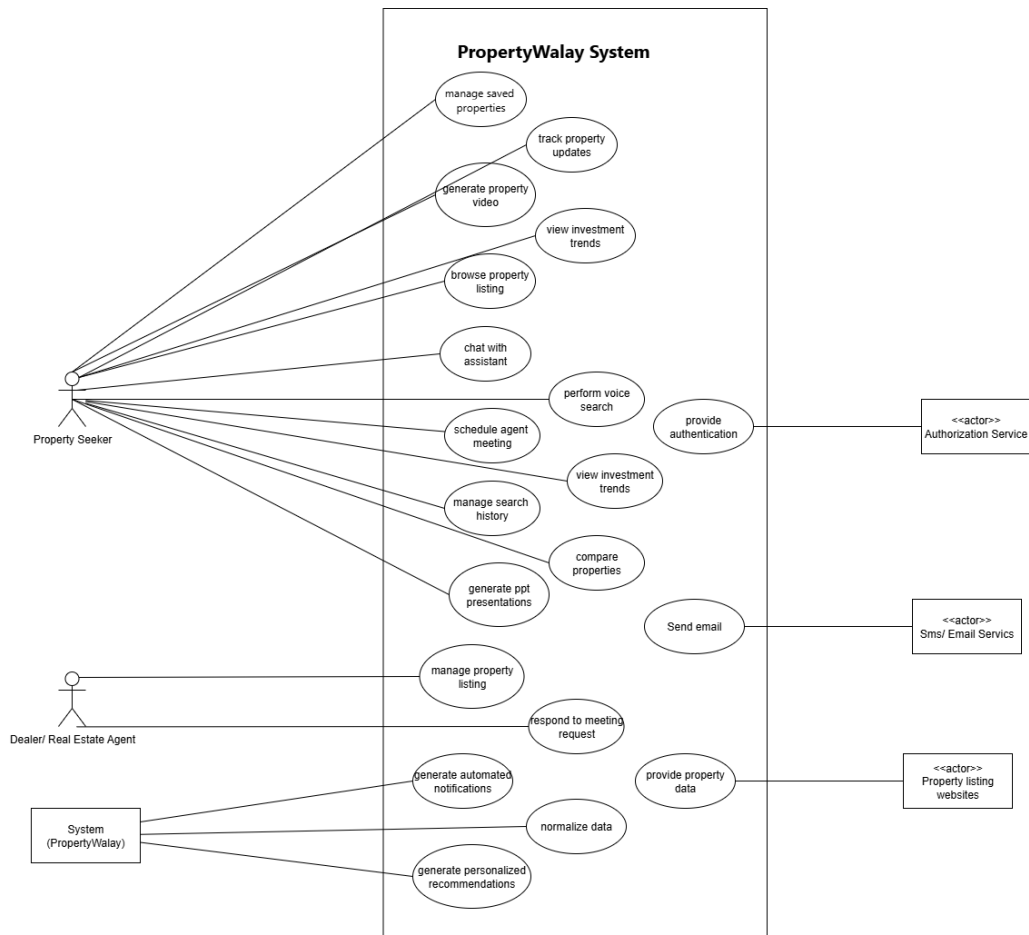


Figure 2.1: Use case diagram for PropertyWalay

The following event response table provides an overview of the designed user interactions with PropertyWalay’s core modules—including Natural Language Understanding, AI-Powered Recommendation Engine, Real-Time Listings Integration, Property Tracking,

Investment Insights, Meeting Automation, and Content Generation—focusing exclusively on user actions against the system. For each user action, the table describes the pre-state, immediate system response and the resulting post-state of the system.

User Action	Pre-Event State	System Response	Post-Event State
Property Seeker chats with assistant	User logged in, chatbot interface displayed	System processes natural language query using NLU, extracts search parameters	Property recommendations displayed based on query
Property Seeker performs voice search	User on chatbot interface, microphone access granted	System converts speech to text, processes query, fetches matching listings	Search results displayed with relevant properties
Property Seeker browses property listing	Search results displayed on screen	System retrieves detailed property information including images, price, location	Property details page loaded with complete information
Property Seeker manages saved properties	User logged in, viewing property details	System adds property to saved list, activates tracking	Property saved, user receives confirmation message
Property Seeker views investment trends	User logged in, on dashboard	System analyzes market data, generates heat maps and trend charts	Investment insights displayed with trending areas highlighted
Property Seeker compares properties	User has viewed multiple properties	System retrieves saved properties, creates comparison table	Side-by-side comparison displayed with key metrics
Property Seeker manages search history	User logged in, on profile page	System retrieves past searches and viewed properties from database	Search history displayed chronologically with quick access
Property Seeker schedules agent meeting	Viewing property details, agent info available	System checks agent availability, presents time slots	Meeting scheduled, confirmation sent to both parties

User Action	Pre-Event State	System Response	Post-Event State
Property Seeker generates PPT presentation	Property details page loaded	System compiles property data, images, location maps into PowerPoint	PPT generated and download link provided
Property Seeker generates property video	Viewing property details page	System creates video from images, adds location overlay and narration	Video generated, preview displayed with download option
Real Estate Agent manages property listing	Agent logged in, on dashboard	System displays agent's active listings with edit options	Listings displayed with management controls
Real Estate Agent responds to meeting request	Meeting request notification received	System displays meeting details and available time slots	Agent confirms meeting, notification sent to property seeker
Real Estate Agent generates automated notifications	New inquiry received for listed property	System creates notification with inquiry details	Agent receives real-time notification via email/SMS
System (Property-Walay) tracks property updates	Property saved by user, listings integration active	System monitors property on source websites for changes	User notified when price changes or property sold
System (Property-Walay) normalizes data	New listings fetched from multiple sources	System standardizes format, removes duplicates, validates data	Clean, normalized data stored in database
System (Property-Walay) generates personalized recommendations	User logged in, sufficient search history available	System applies RAG technology, analyzes preferences	Personalized property recommendations generated
Authorization Service provides authentication	User attempts to login	Service verifies credentials against database	User authenticated and session token generated
Property listing websites provide property data	System requests property listings via API	External websites return property data in JSON/XML format	Fresh property data received and processed

User Action	Pre-Event State	System Response	Post-Event State
SMS/Email Service sends notifications	System triggers notification for price change	Service formats message and sends via email/SMS	User receives notification on registered contact

Table 2.1: Use-case/Event Response Table

2.2 Functional Requirements

This section describes the functional requirements of PropertyWalay organized by modules. Each module contains specific functional requirements that define system capabilities and expected behaviors.

2.2.1 Natural Language Understanding (NLU) Module

Following are the requirements for the Natural Language Understanding module:

1. The system shall accept user queries in English and Roman Urdu without requiring language selection
2. The system shall extract property search parameters (location, price range, property type, size) from natural language text
3. The system shall handle ambiguous queries by asking clarifying questions
4. The system shall support voice-to-text conversion for hands-free property searching
5. The system shall maintain conversation context across multiple query exchanges
6. The system shall handle code-mixing where users combine English and Urdu words in queries

2.2.2 AI-Powered Recommendation Engine

Following are the requirements for the AI-Powered Recommendation module:

1. The system shall generate personalized property recommendations using RAG technology
2. The system shall rank search results based on user preferences, search history, and behavioral patterns
3. The system shall apply collaborative filtering to suggest properties similar users found relevant
4. The system shall refine recommendations based on user feedback (likes, dislikes, saves)
5. The system shall provide explanations for why specific properties are recommended
6. The system shall update recommendation models based on user interactions

2.2.3 Real-Time Listings Integration Module

Following are the requirements for the Real-Time Listings Integration module:

1. The system shall fetch property listings from Zameen.com, Graana.com, and Booking.com
2. The system shall synchronize property data in real-time with update frequency of maximum 1 hour
3. The system shall standardize data formats across different source platforms
4. The system shall validate scraped data for accuracy and completeness
5. The system shall handle API rate limits and implement efficient caching strategies
6. The system shall remove duplicate listings from different sources

2.2.4 Property Tracking and Notification Module

Following are the requirements for the Property Tracking and Notification module:

1. The system shall allow users to save and bookmark properties of interest
2. The system shall monitor saved properties for price changes automatically
3. The system shall send notifications when saved properties are marked as sold or unavailable
4. The system shall track property view counts and engagement metrics over time
5. The system shall provide comparison views showing how saved properties have changed
6. The system shall send notifications via email and SMS based on user preferences

2.2.5 Investment Insights and Trends Module

Following are the requirements for the Investment Insights and Trends module:

1. The system shall analyze listing activity patterns to identify trending neighborhoods
2. The system shall generate heat maps showing property demand across different areas
3. The system shall compare price trends across locations and property types
4. The system shall provide investment scores based on historical data and current trends
5. The system shall create market reports summarizing key insights for different cities
6. The system shall update trend analysis on a weekly basis

2.2.6 Meeting Automation Module

Following are the requirements for the Meeting Automation module:

1. The system shall enable users to request meetings with agents directly through the chatbot
2. The system shall integrate with agent calendars to find available time slots
3. The system shall send automated meeting confirmations to both parties via email and SMS
4. The system shall provide reminders before scheduled meetings (24 hours and 1 hour before)
5. The system shall allow rescheduling and cancellation with automatic notifications
6. The system shall maintain meeting history for both users and agents

2.2.7 Automated Presentation Generation Module

Following are the requirements for the Automated Presentation Generation module:

1. The system shall generate PowerPoint presentations with property details on demand
2. The system shall include property images, floor plans, and location maps in slides
3. The system shall format presentations with professional templates and consistent styling
4. The system shall allow customization of presentation content and branding
5. The system shall enable direct download or email delivery of generated presentations
6. The system shall generate presentations within 30 seconds of request

2.2.8 Location-Based Video Generation Module

Following are the requirements for the Location-Based Video Generation module:

1. The system shall generate short video tours using property images and data
2. The system shall overlay location information and nearby amenities on video
3. The system shall include map visualizations showing property location and surroundings
4. The system shall add voiceover narration describing key property features
5. The system shall enable sharing of generated videos via social media and messaging apps
6. The system shall generate videos within 60 seconds of request

2.2.9 User History and Profile Management Module

Following are the requirements for the User History and Profile Management module:

1. The system shall store user search queries and browsing history across sessions
2. The system shall maintain lists of viewed, saved, and contacted properties

3. The system shall build user preference profiles based on interaction patterns
4. The system shall allow users to review and manage their search history
5. The system shall enable quick access to previously shortlisted properties
6. The system shall allow users to delete search history and clear preferences

2.2.10 Multilingual and Voice Interface Module

Following are the requirements for the Multilingual and Voice Interface module:

1. The system shall support seamless switching between English and Roman Urdu
2. The system shall implement voice recognition for hands-free property searching
3. The system shall convert voice queries to text and process through NLU pipeline
4. The system shall provide text-to-speech responses for accessibility
5. The system shall handle code-mixing where users combine English and Urdu words
6. The system shall support voice commands for common actions (save, compare, contact agent)

2.3 Non-Functional Requirements

This section specifies non-functional requirements that define quality attributes of the PropertyWalay system. These requirements are specific, quantitative, and verifiable.

2.3.1 Performance

1. The system shall provide chatbot responses within 2-3 seconds for 95% of user queries
2. The system shall load property search results within 3 seconds of query submission
3. The system shall generate property recommendations within 4 seconds using RAG technology
4. The system shall support at least 1000 concurrent users without performance degradation
5. The system shall synchronize property listings from external sources within 1 hour intervals
6. The system shall process voice-to-text conversion within 2 seconds of audio input

2.3.2 Reliability

1. The system shall maintain 99% uptime for listing updates and user interactions
2. The system shall have a Mean Time Between Failures (MTBF) of at least 720 hours (30 days)

3. The system shall automatically recover from failures within 5 minutes
4. The system shall maintain data consistency across all integrated property listing sources
5. The system shall implement automatic backup of user data every 24 hours
6. The system shall log all errors and exceptions for debugging and monitoring

2.3.3 Usability

1. The system shall allow property seekers to find and save properties within 3 interactions
2. The system shall provide intuitive chat interface requiring no training for basic operations
3. The system shall display error messages in user-friendly language (English/Roman Urdu)
4. The system shall allow users to retrieve previously viewed properties with a single click
5. The system shall provide contextual help and suggestions during property search
6. The system shall be accessible on mobile devices with responsive design

2.3.4 Security

1. The system shall encrypt all user data at rest using AES-256 encryption
2. The system shall use HTTPS/TLS for all data transmission between client and server
3. The system shall implement user authentication using secure token-based mechanisms
4. The system shall hash and salt all user passwords using bcrypt algorithm
5. The system shall implement rate limiting to prevent brute force attacks (5 failed attempts)
6. The system shall log all authentication attempts and security events
7. The system shall comply with Pakistan's data protection regulations
8. The system shall implement role-based access control for different user types

2.3.5 Scalability

1. The system shall be designed to handle a 300% increase in user load
2. The system shall support horizontal scaling by adding additional server instances
3. The system shall handle at least 10,000 property listings without performance degradation
4. The system shall support expansion to additional property listing sources

2.3.6 Compatibility

1. The system shall be compatible with Chrome, Firefox, Safari, and Edge browsers
2. The system shall support Android (version 8.0+) and iOS (version 12.0+) devices
3. The system shall function properly on screen sizes ranging from 320px to 2560px width
4. The system shall integrate with third-party APIs from Zameen.com, Graana.com, Booking.com

2.3.7 Maintainability

1. The system shall be modular to allow independent updates of individual modules
2. The system code shall follow industry-standard coding conventions and documentation
3. The system shall provide comprehensive API documentation for developers
4. The system shall implement automated testing with at least 80% code coverage

Chapter 3

System Overview

PropertyWalay is an innovative AI-powered chatbot system designed to revolutionize property searching in Pakistan through the integration of natural language understanding, real-time data aggregation, and intelligent recommendation technologies. Key modules include the Natural Language Understanding (NLU) module, which transforms user queries in English and Roman Urdu into structured search parameters; the AI-Powered Recommendation Engine, which personalizes property suggestions using RAG technology; the Real-Time Listings Integration module, which aggregates data from multiple platforms; and the Property Tracking and Notification system, which monitors saved properties for changes. Additional features include automated meeting scheduling with agents, PPT generation for property presentations, and location-based video generation. Developed using a procedural programming approach with a four-tier architecture, PropertyWalay efficiently addresses the challenges of fragmented property search, language barriers, and outdated listings in Pakistan's real estate market. This system not only enhances property search efficiency but also supports users in making informed investment decisions in a dynamic and competitive real estate landscape.

3.1 Architectural Design

The architectural design of PropertyWalay follows a four-tier architecture pattern, providing a clear separation of concerns and enabling scalability, maintainability, and efficient resource management. Each tier is responsible for specific functionalities and communicates with adjacent layers through well-defined interfaces. This modular structure allows independent development, testing, and deployment of components while ensuring seamless integration across the entire system.

3.1.1 Box and Line Diagram

The box and line diagram provides a simplified, high-level view of PropertyWalay’s major subsystems and their interconnections. This initial representation illustrates how different components collaborate to deliver the complete functionality of the property search and recommendation system.

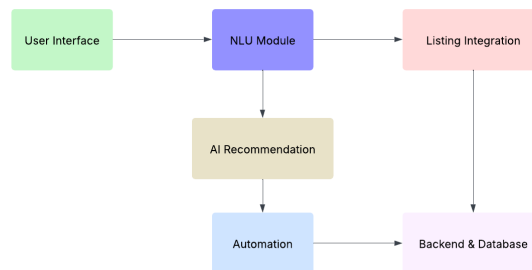


Figure 3.1: Box and Line Diagram of PropertyWalay System

As shown in the diagram, the system is organized around five core components:

User Interface: Serves as the entry point for all user interactions, providing a conversational chatbot interface where property seekers and agents can interact with the system using natural language queries in English or Roman Urdu.

NLU Module: Processes and interprets natural language queries from users, extracting key parameters such as location, property type, price range, and size. This module bridges the gap between human language and system operations.

AI Recommendation: Implements Retrieval-Augmented Generation (RAG) technology to provide personalized property recommendations based on user preferences, search history, and behavioral patterns. This intelligent engine continuously learns and adapts to user needs.

Listings Integration: Aggregates real-time property data from multiple external sources including Zameen.com, Graana.com, and Booking.com. This module ensures data standardization, removes duplicates, and maintains up-to-date property information.

Automation: Handles automated functionalities including property tracking for price changes, meeting scheduling with agents, PPT generation, and video creation. This module reduces manual effort and provides value-added services.

Backend & Database: Manages data persistence, user profiles, search history, saved properties, and system configurations. This foundational layer ensures data integrity and supports all other modules with reliable data storage and retrieval capabilities.

The arrows in the diagram represent data flow and communication pathways between components. The NLU Module acts as the central processing unit, receiving input from the User Interface and coordinating with other modules. Both the Listings Integration and Automation modules interact with the Backend & Database for data operations, while the AI Recommendation module receives processed queries from NLU and retrieves relevant data to generate personalized suggestions.

3.1.2 Four-Tier Architecture

PropertyWalay's architecture is structured into four distinct tiers: Presentation Layer, Application Layer, Integration Layer, and Data Layer. This layered approach provides modularity, scalability, and clear separation of responsibilities across the system.

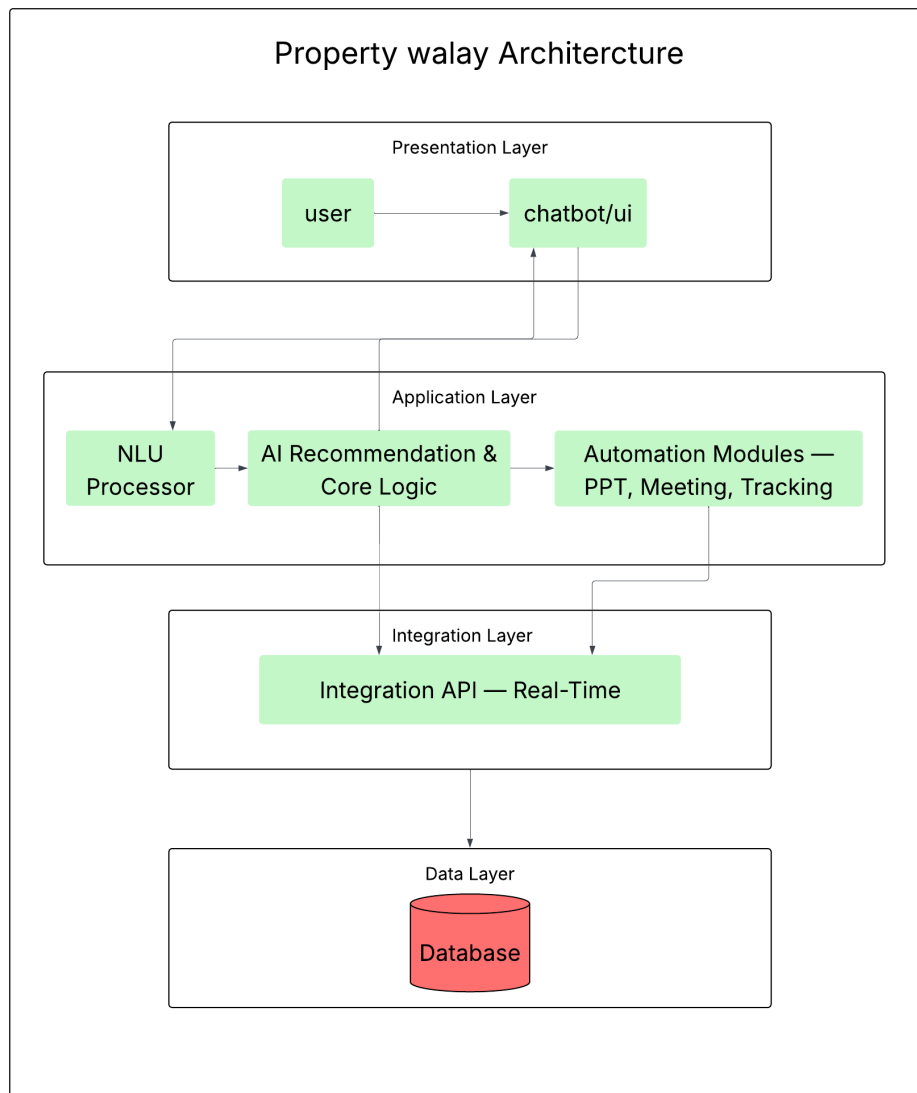


Figure 3.2: Four-Tier Architecture of PropertyWalay System

3.1.2.1 Presentation Layer

The Presentation Layer serves as the user-facing interface of PropertyWalay, providing an intuitive and accessible chatbot-based user interface. This layer is responsible for:

- **User Interaction:** Facilitating user input through text and voice-based queries in English and Roman Urdu
- **Chatbot Interface:** Providing a conversational interface that mimics natural human interaction for property searching
- **Response Rendering:** Displaying property listings, recommendations, comparisons, investment insights, and generated content (PPTs, videos)
- **Session Management:** Maintaining user sessions and conversation context across multiple interactions

The Presentation Layer is implemented using ReactJS for web interfaces and is designed to be responsive across desktop and mobile devices. It communicates bidirectionally with the Application Layer to send user requests and receive system responses.

3.1.2.2 Application Layer

The Application Layer contains the core business logic and intelligent processing capabilities of PropertyWalay. This tier comprises three primary modules:

NLU Processor: Receives raw natural language queries from the Presentation Layer and performs:

- Natural language parsing and intent recognition
- Entity extraction (location, price, property type, size)
- Language detection and processing (English/Roman Urdu)
- Query contextualization based on conversation history

AI Recommendation & Core Logic: Implements the intelligent recommendation engine using RAG technology:

- Retrieval of relevant property listings from the database
- Application of user preference models and behavioral patterns
- Ranking and filtering of properties based on multiple criteria

- Generation of personalized recommendations and explanations
- Investment trend analysis and insights generation

Automation Modules: Manages automated features that enhance user experience:

- **PPT Generation:** Compiles property data, images, and location maps into professional PowerPoint presentations
- **Meeting Scheduling:** Coordinates meetings between property seekers and agents by checking availability and sending confirmations
- **Property Tracking:** Monitors saved properties for price changes and availability status, triggering notifications when updates occur
- **Video Generation:** Creates location-based property videos with overlays and narration

The Application Layer orchestrates the flow of information between user requests and data operations, implementing the core intelligence and automation that distinguishes PropertyWalay from traditional property search platforms.

3.1.2.3 Integration Layer

The Integration Layer acts as a bridge between the application logic and external data sources, handling real-time data synchronization and API communication:

Integration API — Real-Time: This component is responsible for:

- Connecting to external property listing platforms (Zameen.com, Graana.com, Booking.com) via APIs or web scraping
- Fetching real-time property data at regular intervals (maximum 1-hour refresh rate)
- Data transformation and standardization across different source formats
- Duplicate detection and removal across multiple sources
- Error handling and retry mechanisms for failed API calls
- Rate limiting compliance to respect external API constraints

This layer ensures that PropertyWalay always presents users with the most current property information available, addressing one of the major pain points in Pakistan's real estate market—outdated listings.

3.1.2.4 Data Layer

The Data Layer provides persistent storage and data management capabilities for the entire system:

Database: The database component manages:

- **User Data:** User profiles, authentication credentials, preferences, and roles (property seekers, agents, administrators)
- **Property Listings:** Aggregated and standardized property data from multiple sources with timestamps
- **Search History:** Records of user queries, viewed properties, and search patterns
- **Saved Properties:** User-saved property lists with tracking status
- **Meeting Records:** Scheduled meetings between seekers and agents with status updates
- **System Logs:** Transaction logs, error logs, and audit trails
- **Cache Data:** Frequently accessed data for improved performance

PropertyWalay uses a combination of PostgreSQL for structured relational data (users, meetings, transactions) and MongoDB for semi-structured data (property listings, search history, user preferences), providing flexibility and optimal performance for different data types.

3.1.3 Architecture Benefits

The four-tier architecture of PropertyWalay offers several key advantages:

- **Modularity:** Each tier can be developed, tested, and deployed independently
- **Scalability:** Individual tiers can be scaled horizontally based on load (e.g., adding more Integration Layer instances during peak scraping times)
- **Maintainability:** Clear separation of concerns makes the system easier to maintain and update
- **Security:** Layered architecture enables implementation of security measures at each tier
- **Flexibility:** New features or external integrations can be added without affecting other layers

- **Performance:** Each layer can be optimized independently for its specific responsibilities

Overall, the system follows a procedural approach within a four-tier architecture, where different services are responsible for specific tasks or functionalities. These services work together to provide a modular, scalable, and flexible system architecture tailored to the unique requirements of Pakistan's real estate market.

3.2 Design Models

3.2.1 Activity Diagram

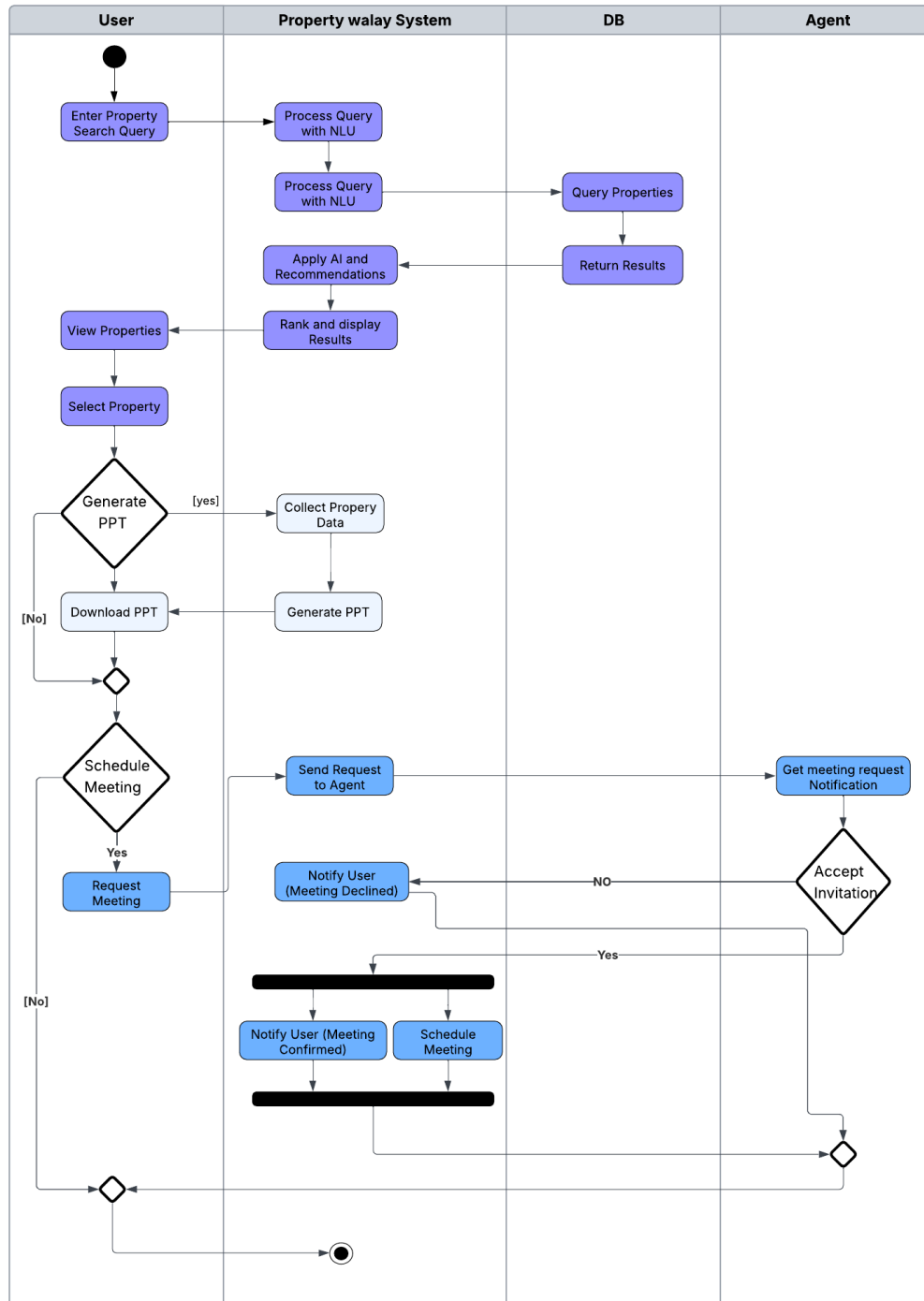


Figure 3.3: Activity Diagram of PropertyWalay System

The activity diagram of PropertyWalay visually represents the dynamic flow of operations, capturing the sequence of activities and decision points within the platform's modules across pertinent user roles. This activity diagram illustrates the complete user journey from entering a property search query through to potentially scheduling a meeting with a real estate agent.

The diagram is organized into four swim lanes representing the major actors and system components: **User**, **Property Walay System**, **DB (Database)**, and **Agent**.

3.2.1.1 Property Search Flow

The activity flow begins when the user enters a property search query through the chatbot interface. The system processes this query using the NLU module to extract search parameters such as location, property type, price range, and size. The processed query is then sent to the database which queries the aggregated property listings from multiple sources.

After retrieving relevant properties from the database, the system applies AI-powered recommendations using RAG technology. This step analyzes user preferences, search history, and behavioral patterns to personalize the results. The recommendation engine ranks and filters properties based on multiple criteria before displaying them to the user.

3.2.1.2 Property Selection and PPT Generation

Once the user views the property recommendations, they can select a specific property for more detailed information. At this point, the user is presented with a decision: whether to generate a PowerPoint presentation for the selected property.

If the user chooses **yes** for PPT generation, the system collects comprehensive property data including images, floor plans, location maps, price details, and amenities. The system then generates a professionally formatted PowerPoint presentation and provides a download link to the user. This feature is particularly useful for users who want to share property details with family members or for agents preparing for client meetings.

If the user chooses **no**, the system proceeds directly to the next decision point.

3.2.1.3 Meeting Scheduling Flow

After handling the PPT generation decision, the user is presented with another option: whether to schedule a meeting with the property agent. This decision point enables direct interaction between property seekers and real estate agents.

If the user selects **no** for scheduling a meeting, the activity flow terminates, and the user can continue browsing other properties or exit the system.

If the user selects **yes**, the following sequence occurs:

1. The user formally requests a meeting through the interface
2. The system sends a meeting request to the designated agent
3. The agent receives a meeting request notification
4. The agent must decide whether to accept the meeting invitation

3.2.1.4 Agent Response Handling

The agent's response to the meeting invitation creates two possible paths:

Meeting Declined: If the agent chooses **no**, the system notifies the user that the meeting has been declined. The user receives this notification and the process terminates. This ensures transparency and allows the user to either contact a different agent or reconsider their options.

Meeting Accepted: If the agent chooses **yes** to accept the meeting invitation, a parallel process is triggered where the system performs two actions simultaneously:

- Notifies the user that the meeting has been confirmed
- Schedules the meeting with date, time, and location details

Both notifications are sent concurrently to ensure both parties are immediately informed of the confirmed meeting. After these notifications are delivered, the activity flow reaches its final termination point.

3.2.1.5 Key Decision Points

The activity diagram highlights three critical decision points in the user journey:

1. **Generate PPT?** — Determines whether the user wants a downloadable presentation
2. **Schedule Meeting?** — Determines whether the user wants direct agent interaction
3. **Accept Invitation?** — Determines whether the agent is available and willing to meet

These decision points provide flexibility and user control throughout the property search and engagement process.

3.2.1.6 System Benefits Illustrated

The activity diagram effectively demonstrates several key features of PropertyWalay:

- **Automated Intelligence:** The NLU processing and AI recommendation steps show how the system intelligently interprets and responds to user needs
- **Content Generation:** The PPT generation flow illustrates value-added automation features
- **Agent Coordination:** The meeting scheduling sequence demonstrates how PropertyWalay facilitates communication between seekers and agents
- **User Empowerment:** Multiple decision points give users control over their experience
- **Real-time Integration:** Database queries reflect the system's access to aggregated, up-to-date property listings

This comprehensive activity flow ensures that PropertyWalay provides a seamless, intelligent, and user-centric property search experience from initial query to potential agent meeting, addressing the fragmented nature of traditional property search processes in Pakistan.

3.2.2 Data Flow Diagram

Data Flow Diagrams (DFDs) provide a visual representation of how data moves through the PropertyWalay system, illustrating the flow of information between processes, data stores, and external entities. The hierarchical approach—from Context Diagram (Level 0) to increasingly detailed levels—enables stakeholders to understand both the high-level system architecture and the intricate operational details of each module.

3.2.2.1 DFD Level 0 - Context Diagram

Level 0 Data Flow Diagram provides a top-level, simplified representation of PropertyWalay's data flow. It offers a broad overview of the system's architecture, illustrating how data moves between the system and end users (Property Seekers, Real Estate Agents, and External Property Listing Platforms), setting the stage for more detailed analysis in lower-level DFDs.

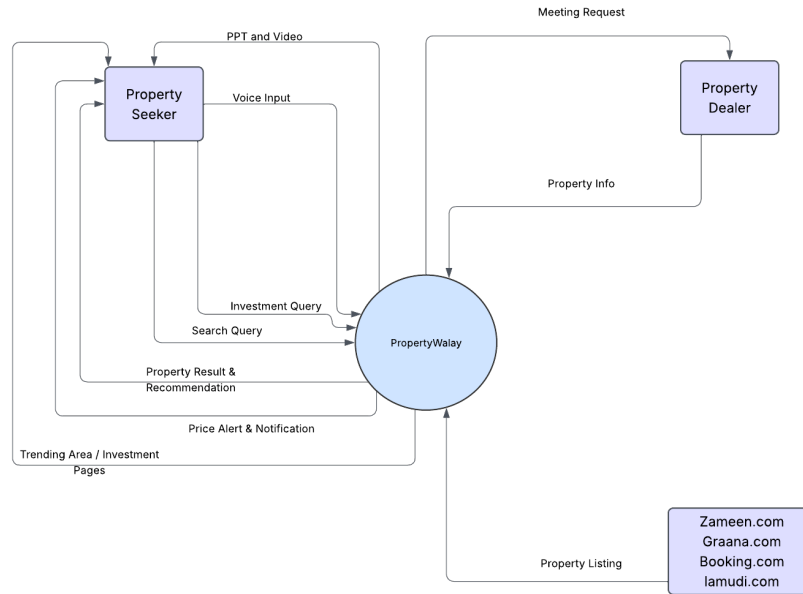


Figure 3.4: Context Diagram of the PropertyWalay System

The Context Diagram shows PropertyWalay as a single process (circle in the center) interacting with three primary external entities:

- **Property Seeker:** Sends search queries, voice input, investment queries, and meeting requests to the system. Receives property results, recommendations, price alerts, notifications, trending investment pages, and generated content (PPTs/Videos).
- **Property Dealer/Agent:** Receives meeting requests and property information from the system. Sends meeting confirmations and property details back to PropertyWalay.
- **External Property Platforms (Zameen.com, Graana.com, Booking.com, Lamudi.com):** Provide property listings to PropertyWalay through API calls and web scraping.

This high-level view establishes the system boundary and identifies the major information flows without detailing internal processes.

3.2.2.2 DFD Level 1

Level 1 Data Flow Diagram delves deeper into the details of PropertyWalay, breaking down the system into more specific subprocesses. Data stores, external entities, and data flows are further detailed at this level, providing a more comprehensive understanding of how data is processed and exchanged at a more specific operational level. The Level 1 DFD acts as an intermediary step between the high-level abstraction of the Level 0 diagram and the more intricate, lower-level DFDs that follow, offering a balanced level of

detail for system analysis and design.

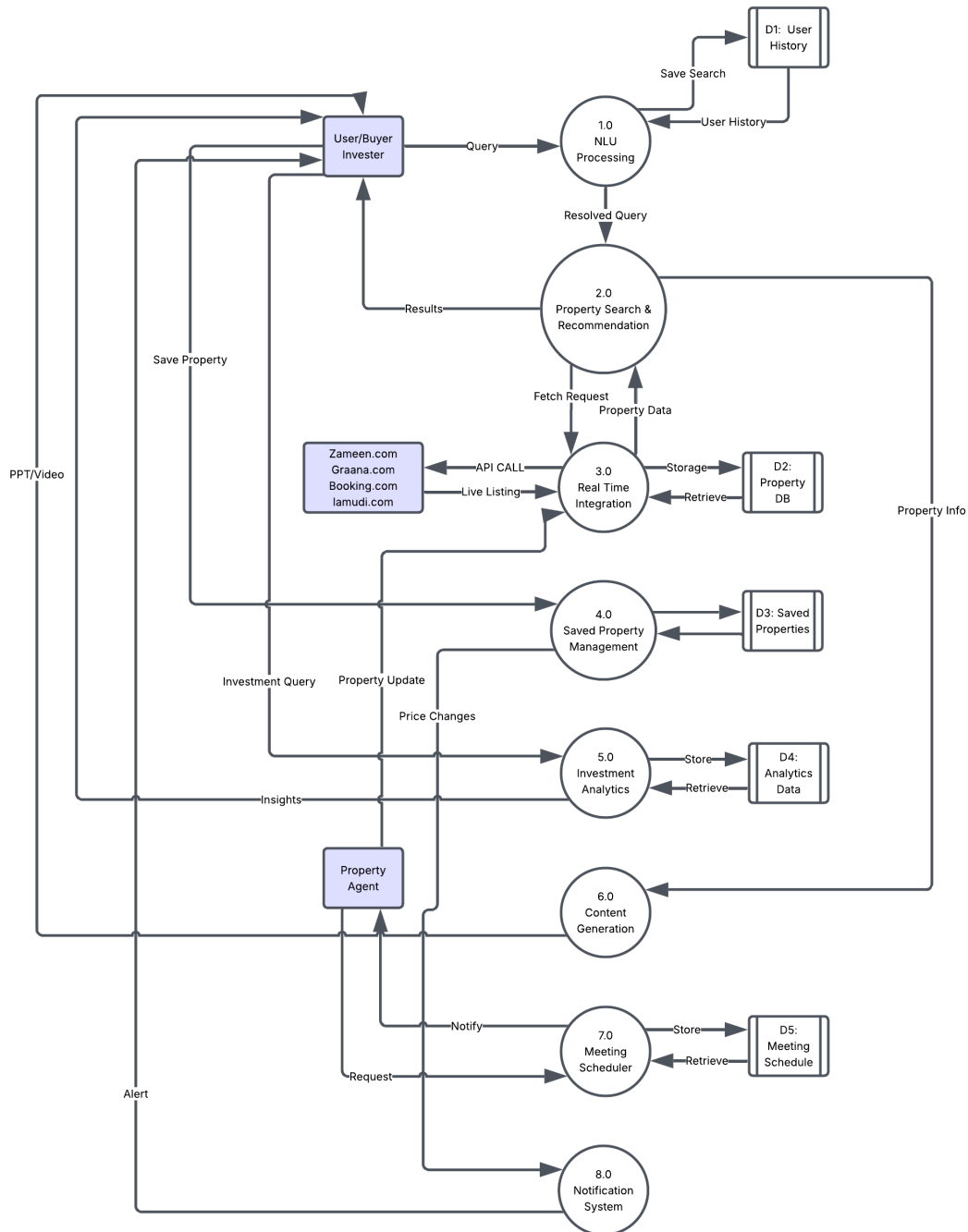


Figure 3.5: DFD Level 1 Diagram of the PropertyWalay System

The Level 1 DFD decomposes PropertyWalay into eight major processes:

Process 1.0 - NLU Processing:

- **Input:** Raw queries from User/Buyer/Investor
- **Processing:** Natural language understanding, entity extraction, language detection

- **Output:** Resolved query with structured parameters
- **Data Store Interaction:** Saves search to D1: User History, retrieves user history for context

Process 2.0 - Property Search & Recommendation:

- **Input:** Resolved query from Process 1.0
- **Processing:** RAG-based recommendation engine, collaborative filtering, property ranking
- **Output:** Personalized property results to user
- **Data Store Interaction:** Fetches property data from D2: Property DB

Process 3.0 - Real Time Integration:

- **Input:** Fetch requests from Process 2.0, API calls to external platforms
- **Processing:** Web scraping, data normalization, duplicate removal
- **Output:** Live listings to Process 2.0
- **Data Store Interaction:** Stores/retrieves property info from D2: Property DB
- **External Entity:** Zameen.com, Graana.com, Booking.com, Lamudi.com

Process 4.0 - Saved Property Management:

- **Input:** Save property requests from user
- **Processing:** Store saved properties, track changes
- **Output:** Property update alerts to Process 8.0
- **Data Store Interaction:** Stores/retrieves from D3: Saved Properties

Process 5.0 - Investment Analytics:

- **Input:** Investment queries from user, price changes from Process 4.0
- **Processing:** Trend analysis, heat map generation, market insights
- **Output:** Investment insights to user
- **Data Store Interaction:** Stores/retrieves analytics from D4: Analytics Data

Process 6.0 - Content Generation:

- **Input:** PPT/Video requests from user, property info from D2
- **Processing:** PowerPoint assembly, video compilation with maps and narration
- **Output:** Generated PPT/Video content to user
- **Data Store Interaction:** Retrieves property data from D2: Property DB

Process 7.0 - Meeting Scheduler:

- **Input:** Meeting requests from user, confirmations from Property Agent
- **Processing:** Schedule coordination, calendar integration, reminder generation
- **Output:** Notifications to user via Process 8.0, meeting requests to agent

- **Data Store Interaction:** Stores/retrieves from D5: Meeting Schedule

Process 8.0 - Notification System:

- **Input:** Alerts from Process 4.0, notifications from Process 7.0
- **Processing:** Notification formatting, delivery channel selection (email/SMS)
- **Output:** Price alerts, meeting confirmations, property updates to user

Data Stores:

- **D1: User History** - Stores search queries, viewed properties, user preferences
- **D2: Property DB** - Central repository for aggregated property listings
- **D3: Saved Properties** - User-bookmarked properties with tracking status
- **D4: Analytics Data** - Price trends, view counts, market insights
- **D5: Meeting Schedule** - Agent meeting records and calendar data

3.2.2.3 DFD Level 2 - AI-Powered Property Recommendation

This Level 2 diagram provides a detailed breakdown of PropertyWalay's "Property Search & Recommendation" (Process 2.0) operations, offering a granular perspective on the flow of data and functionalities. The recommendation process is further dissected into "Query Parser", "Retrieve User Profile", "Filter Properties", "RAG-based Recommend Engine", "Format Response", and "Update Search Log" which are the key components for the AI Recommendation Module.

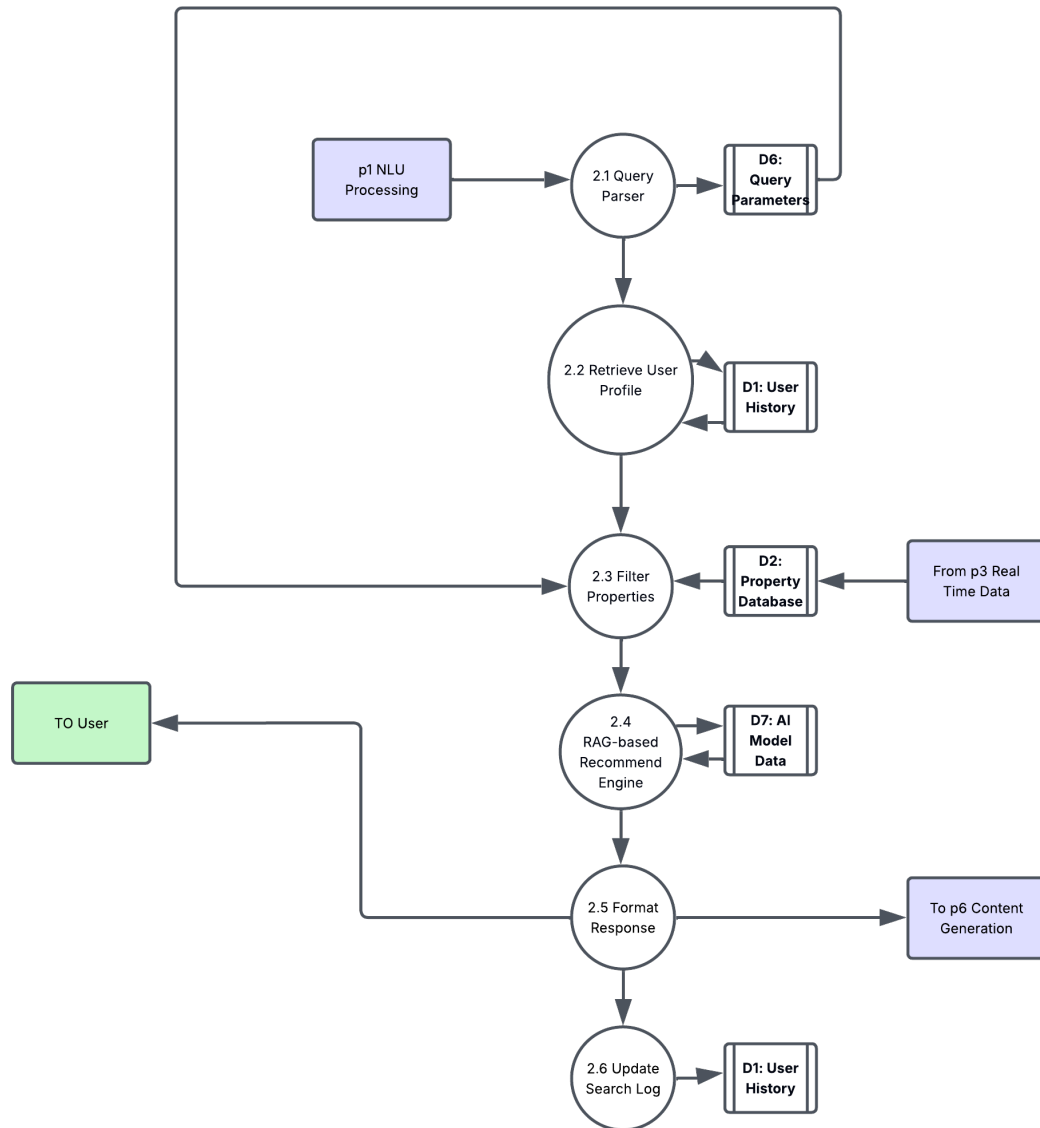


Figure 3.6: DFD Level 2 Diagram of AI-Powered Property Recommendation

Process 2.1 - Query Parser:

- **Input:** Resolved query from p1 NLU Processing
- **Processing:** Extract structured parameters (location, price, beds, baths, property type)
- **Output:** Query parameters to Process 2.2 and D6: Query Parameters

Process 2.2 - Retrieve User Profile:

- **Input:** User identification from Process 2.1
- **Processing:** Fetch search history, saved properties, behavioral patterns
- **Output:** User profile data to Process 2.3

- **Data Store Interaction:** Retrieves from D1: User History

Process 2.3 - Filter Properties:

- **Input:** Query parameters from Process 2.1, user profile from Process 2.2
- **Processing:** Apply filters based on price, location, size, amenities
- **Output:** Filtered property set to Process 2.4
- **Data Store Interaction:** Retrieves from D2: Property Database and p3 Real Time Data

Process 2.4 - RAG-based Recommend Engine:

- **Input:** Filtered properties from Process 2.3
- **Processing:** Retrieval-Augmented Generation, collaborative filtering, ranking algorithm
- **Output:** Ranked recommendations to Process 2.5
- **Data Store Interaction:** Retrieves from D7: AI Model Data

Process 2.5 - Format Response:

- **Input:** Ranked recommendations from Process 2.4
- **Processing:** Format results for display, prepare explanations
- **Output:** Formatted property listings to User and to p6 Content Generation

Process 2.6 - Update Search Log:

- **Input:** Query parameters and results from Process 2.5
- **Processing:** Log search activity, update user profile
- **Data Store Interaction:** Updates D1: User History

3.2.2.4 DFD Level 2 - Content Generation (PPT & Video)

This Level 2 diagram provides a detailed breakdown of PropertyWalay's "Content Generation" (Process 6.0) operations, offering a granular perspective on the flow of data and functionalities. The content generation is further dissected into "Validate Request", "Fetch Property Details", "Retrieve Images", "Generate Map & Location", "Select Template", "Compose PPT Slides", "Generate Video Tour", "Quality Check & Export", and "Store Content History" which are the key components for the Automated Presentation Generation and Video Generation modules.

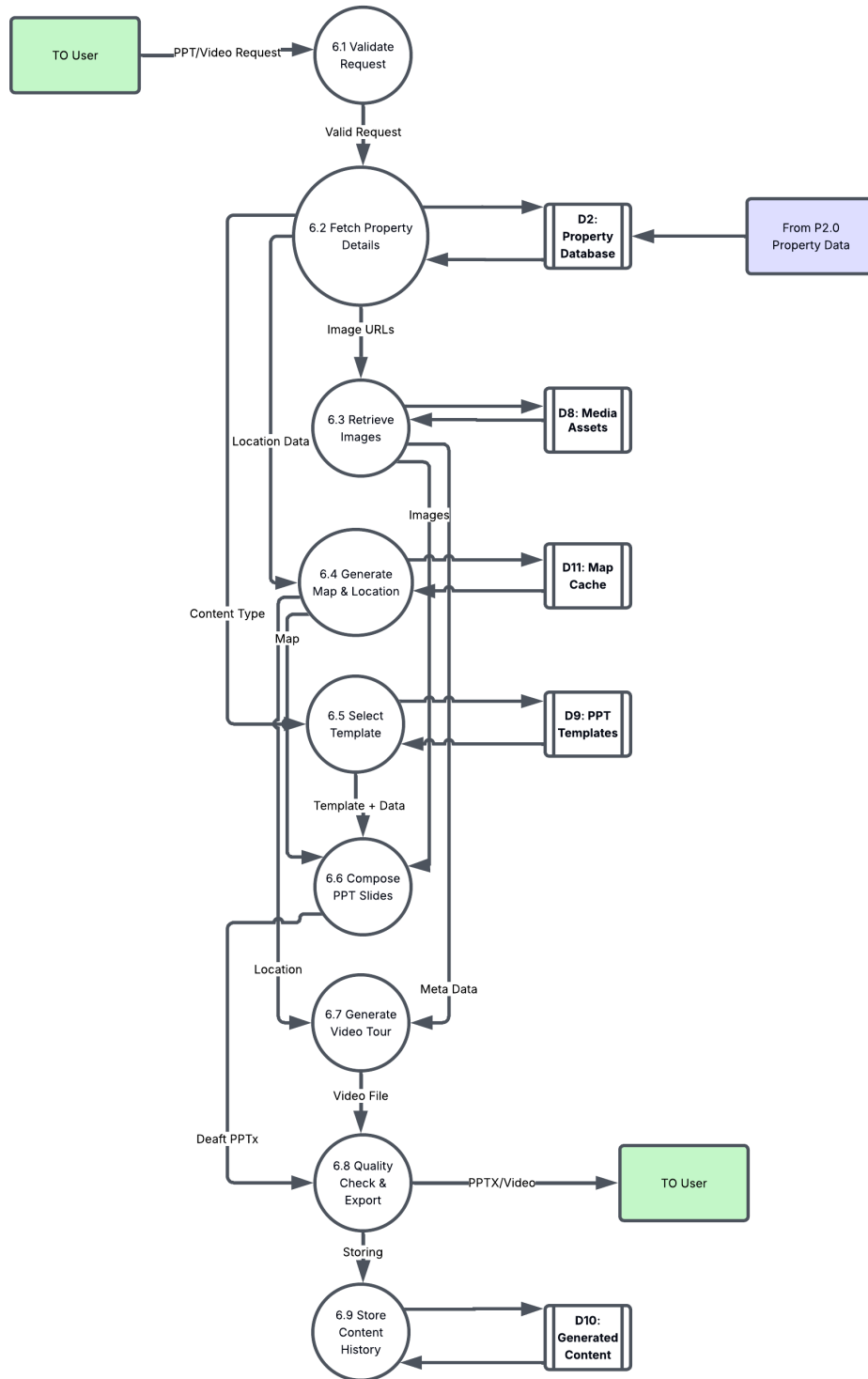


Figure 3.7: DFD Level 2 Diagram of Content Generation (PPT & Video)

Process 6.1 - Validate Request:

- **Input:** PPT/Video request from TO User
- **Processing:** Verify property ID, check user permissions, validate content type

- **Output:** Valid request confirmation to Process 6.2

Process 6.2 - Fetch Property Details:

- **Input:** Validated property ID from Process 6.1
- **Processing:** Retrieve comprehensive property information
- **Output:** Property data to Process 6.3, Image URLs to Process 6.3
- **Data Store Interaction:** Retrieves from D2: Property Database
- **External Input:** From P2.0 Property Data

Process 6.3 - Retrieve Images:

- **Input:** Image URLs from Process 6.2
- **Processing:** Fetch images from storage, optimize for presentation
- **Output:** Images to Process 6.4 and Process 6.6, Location data to Process 6.4
- **Data Store Interaction:** Retrieves from D8: Media Assets

Process 6.4 - Generate Map & Location:

- **Input:** Location data from Process 6.3, Images from Process 6.3
- **Processing:** Create location maps, identify nearby amenities, generate map overlays
- **Output:** Map to Process 6.5 and Process 6.6, Content type decision to Process 6.5
- **Data Store Interaction:** Caches maps in D11: Map Cache

Process 6.5 - Select Template:

- **Input:** Content type from Process 6.4, Map from Process 6.4
- **Processing:** Choose appropriate PPT template based on property type
- **Output:** Template + Data to Process 6.6
- **Data Store Interaction:** Retrieves from D9: PPT Templates

Process 6.6 - Compose PPT Slides:

- **Input:** Template + Data from Process 6.5, Images from Process 6.4, Map from Process 6.4
- **Processing:** Assemble slides with property details, images, pricing, amenities
- **Output:** Draft PPTx to Process 6.8, Location to Process 6.7

Process 6.7 - Generate Video Tour:

- **Input:** Location from Process 6.6, Meta data from Process 6.6
- **Processing:** Compile images into video, add location overlays, generate voiceover narration
- **Output:** Video file to Process 6.8

Process 6.8 - Quality Check & Export:

- **Input:** Draft PPTx from Process 6.6, Video file from Process 6.7
- **Processing:** Validate formatting, check file integrity, compress for delivery
- **Output:** Final PPTx/Video to TO User, Storing command to Process 6.9

Process 6.9 - Store Content History:

- **Input:** Storing command from Process 6.8
- **Processing:** Log generation metadata, track user downloads
- **Data Store Interaction:** Stores in D10: Generated Content

3.2.2.5 DFD Level 2 - Meeting Automation

This Level 2 diagram provides a detailed breakdown of PropertyWalay's "Meeting Scheduler" (Process 7.0) operations, offering a granular perspective on the flow of data and functionalities. The meeting automation is further dissected into "Receive Request", "Validate User & Property", "Check Agent Availability", "Find Available Time Slot", "Create Meeting", "Notify Agent", "Send User Confirmation", "Generate Reminders", and "Update Status" which are the key components for the Meeting Automation Module.

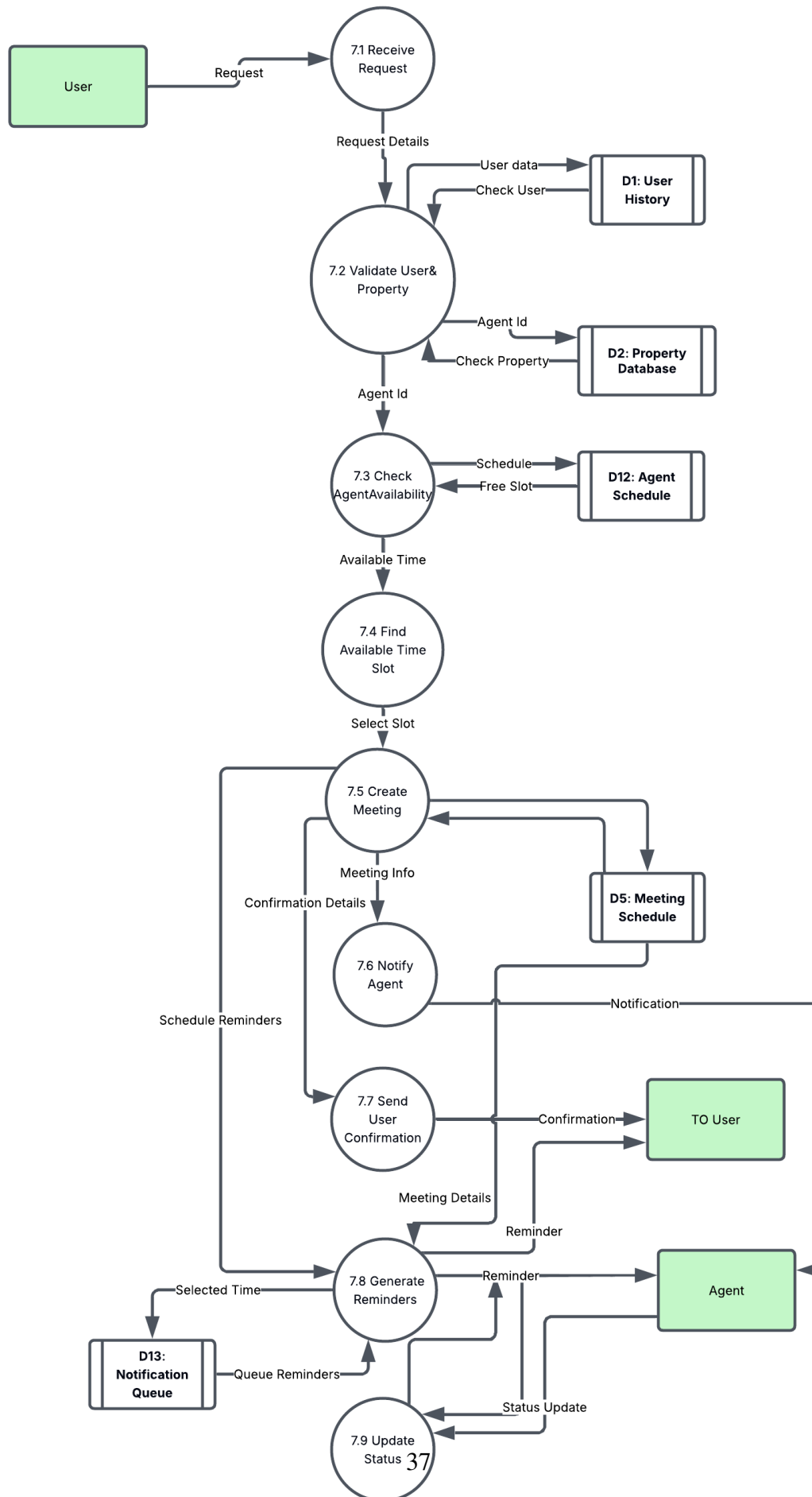


Figure 3.8: DFD Level 2 Diagram of Meeting Automation

Process 7.1 - Receive Request:

- **Input:** Meeting request from User
- **Processing:** Parse request details, extract property and user information
- **Output:** Request details to Process 7.2

Process 7.2 - Validate User & Property:

- **Input:** Request details from Process 7.1
- **Processing:** Verify user authentication, check property exists, validate agent assignment
- **Output:** User data to Process 7.3, Agent ID to Process 7.3
- **Data Store Interaction:** Checks D1: User History and D2: Property Database

Process 7.3 - Check Agent Availability:

- **Input:** Agent ID from Process 7.2
- **Processing:** Query agent calendar, identify free time slots
- **Output:** Schedule/Free Slot to Process 7.4, Agent ID to Process 7.4
- **Data Store Interaction:** Retrieves from D12: Agent Schedule

Process 7.4 - Find Available Time Slot:

- **Input:** Schedule/Free Slot from Process 7.3, Agent ID from Process 7.3
- **Processing:** Match user preferences with agent availability, propose time slots
- **Output:** Available time to Process 7.5, Select slot confirmation to Process 7.5

Process 7.5 - Create Meeting:

- **Input:** Available time and Select slot from Process 7.4
- **Processing:** Generate meeting record, assign unique meeting ID, set meeting parameters
- **Output:** Meeting info to Process 7.6 and Process 7.7, Confirmation details to Process 7.7
- **Data Store Interaction:** Stores in D5: Meeting Schedule

Process 7.6 - Notify Agent:

- **Input:** Meeting info from Process 7.5
- **Processing:** Format notification with meeting details, send to agent
- **Output:** Notification to Agent

Process 7.7 - Send User Confirmation:

- **Input:** Meeting info and Confirmation details from Process 7.5
- **Processing:** Generate confirmation message with meeting details, date, time, location

- **Output:** Confirmation to TO User, Meeting details to Process 7.8, Schedule reminders command to Process 7.8

Process 7.8 - Generate Reminders:

- **Input:** Meeting details from Process 7.7, Schedule reminders command from Process 7.7, Selected time
- **Processing:** Create reminder schedule (24 hours before, 1 hour before), queue reminders
- **Output:** Reminder to TO User, Reminder to Agent, Status update to Process 7.9
- **Data Store Interaction:** Stores in D13: Notification Queue

Process 7.9 - Update Status:

- **Input:** Status update from Agent, Status update from Process 7.8
- **Processing:** Track meeting status (scheduled, completed, cancelled, rescheduled)
- **Data Store Interaction:** Updates D5: Meeting Schedule

These detailed Level 2 DFDs provide comprehensive insights into the internal workings of PropertyWalay's major subsystems, demonstrating the intricate data transformations and interactions that enable intelligent property search, automated content generation, and seamless agent coordination. Each process is carefully designed to support the system's functional requirements while maintaining data integrity and optimal performance.

3.2.3 System-level Sequence Diagram

This section contains system sequence diagrams for the PropertyWalay project. As the system operates autonomously, only the input and output are depicted. The system is treated as a black box, adhering to system sequence diagram rules.

3.2.3.1 Process Natural Language Query

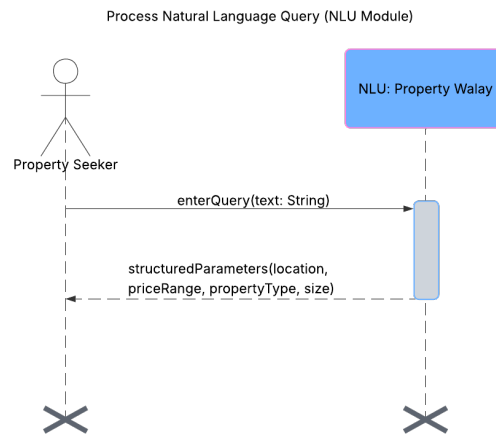


Figure 3.9: System Sequence Diagram for Process Natural Language Query function (NLU Module)

3.2.3.2 Search Properties

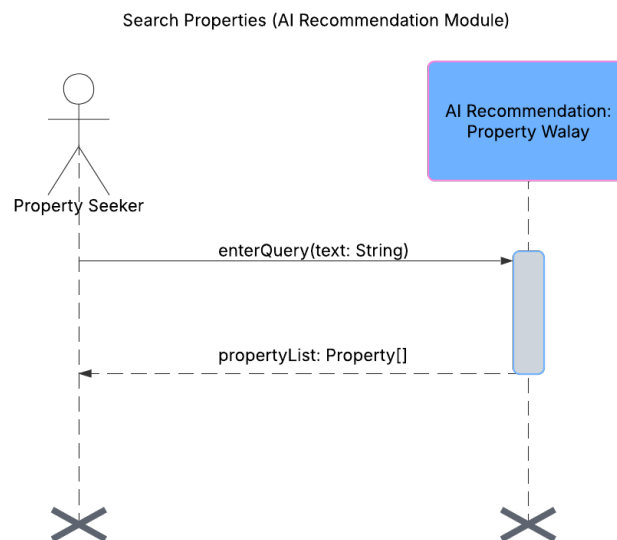


Figure 3.10: System Sequence Diagram for Search Properties function (AI Recommendation Module)

3.2.3.3 View Property Details

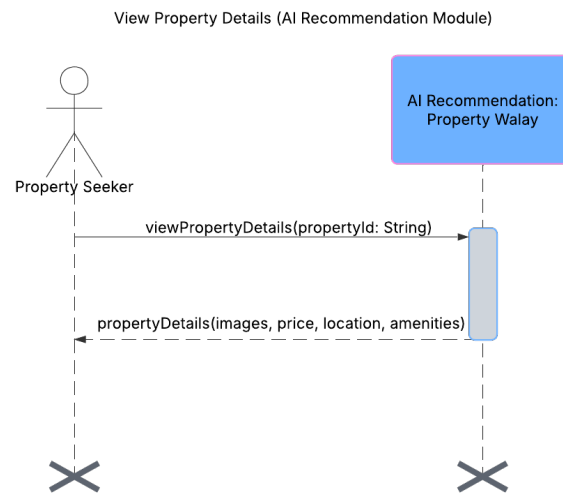


Figure 3.11: System Sequence Diagram for View Property Details function (Database Module)

3.2.3.4 Save Property

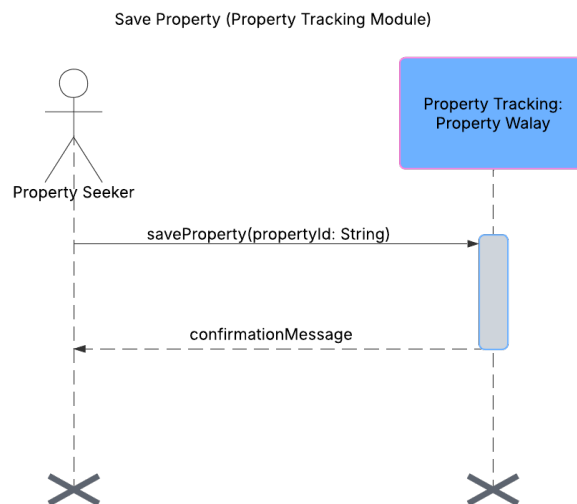


Figure 3.12: System Sequence Diagram for Save Property function (Database Module)

3.2.3.5 Get Saved Properties

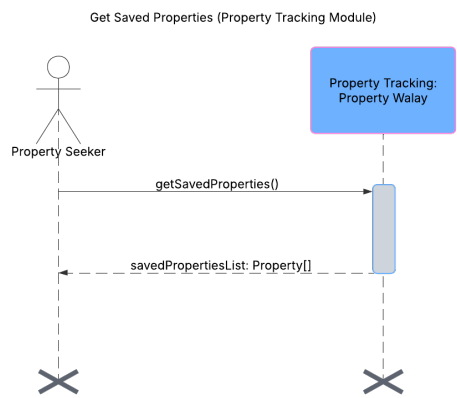


Figure 3.13: System Sequence Diagram for Get Saved Properties function (Database Module)

3.2.3.6 Generate PPT

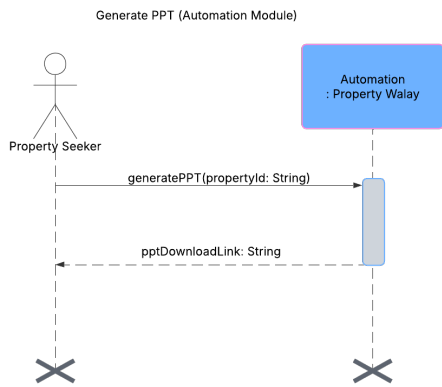


Figure 3.14: System Sequence Diagram for Generate PPT function (Automation Module)

3.2.3.7 Generate Property Video

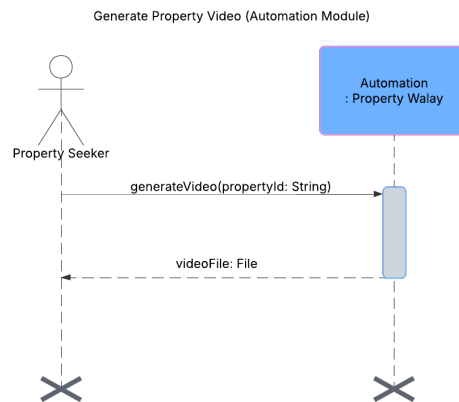


Figure 3.15: System Sequence Diagram for Generate Property Video function (Automation Module)

3.2.3.8 Request Agent Meeting

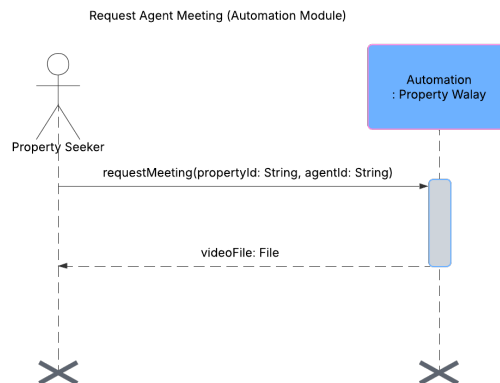


Figure 3.16: System Sequence Diagram for Request Agent Meeting function (Automation Module)

3.2.3.9 Agent Accepts Meeting

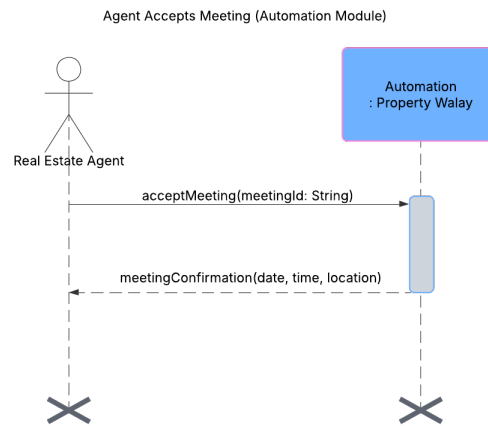


Figure 3.17: System Sequence Diagram for Agent Accepts Meeting function (Automation Module)

3.2.3.10 Get Search History

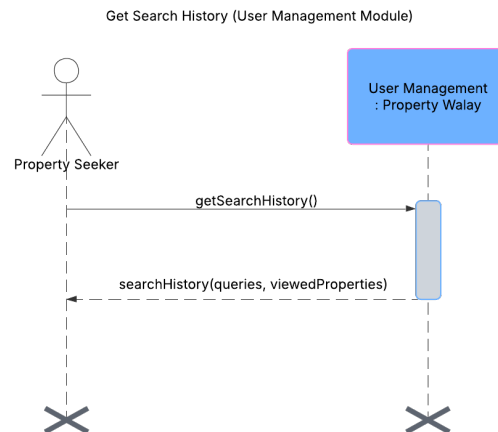


Figure 3.18: System Sequence Diagram for Get Search History function (Database Module)

3.2.3.11 Compare Properties

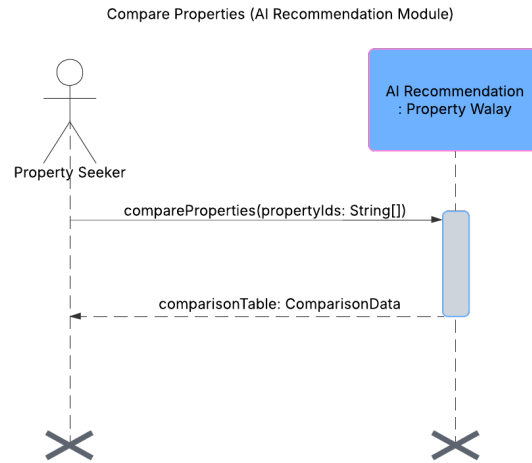


Figure 3.19: System Sequence Diagram for Compare Properties function (AI Recommendation Module)

3.2.3.12 Get Investment Insights

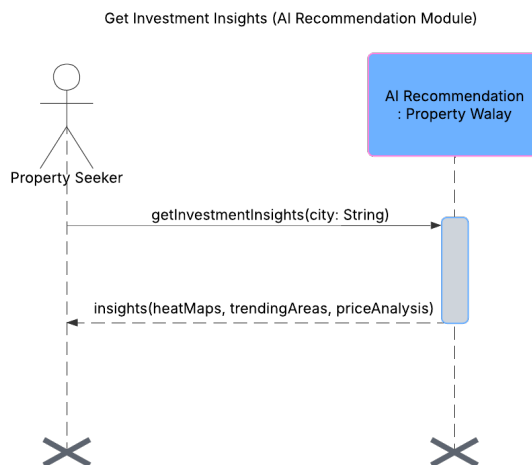


Figure 3.20: System Sequence Diagram for Get Investment Insights function (AI Recommendation Module)

3.2.3.13 Track Property Updates

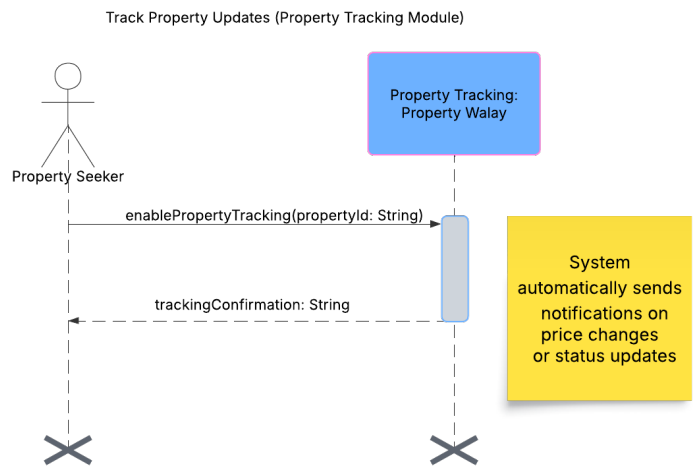


Figure 3.21: System Sequence Diagram for Track Property Updates function (Automation Module)

3.2.3.14 Fetch Listings from External Source

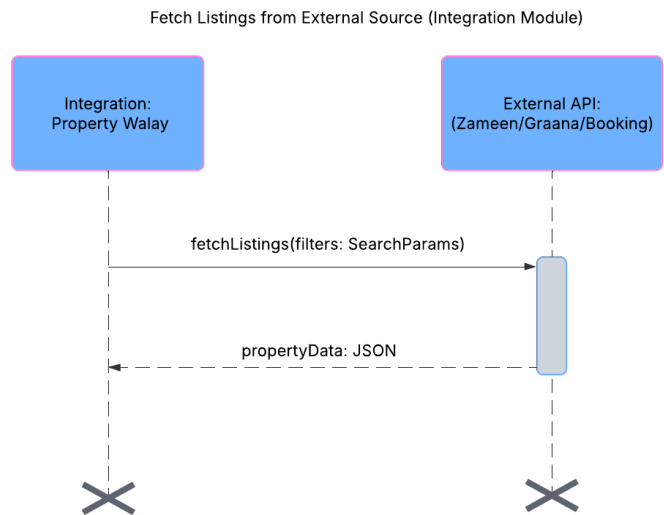


Figure 3.22: System Sequence Diagram for Fetch Listings from External Source function (Integration Module)

3.3 Data Design

When it comes to representing intricate data structures like database schemas, sometimes the most effective method is the simplest: presenting it in its raw, textual form. While there are various visualization tools and methods available for database structures or flow diagrams, a database schema's detailed nature often means that the clearest way to understand it is by viewing the actual text. This ensures that none of the schema's nuances are lost in translation.

The PropertyWalay database schema has been meticulously designed using **Supabase (PostgreSQL)** to capture essential property information, user interactions, and system operations in a structured manner. The schema utilizes UUID-based primary keys ensuring global uniqueness and facilitating data consistency across distributed operations. The `our_id` attribute in the properties table, while serving as a unique identifier, is also pivotal for referencing properties in other tables, ensuring data consistency and facilitating relational database operations. This interconnectedness is vital for maintaining a holistic view of property listings, user preferences, and price tracking across the system.

3.3.1 Properties Table

The properties table serves as the central repository for all property listings aggregated from external sources (Graana, Lamudi, and Zameen). The `our_id` attribute uniquely identifies each property within PropertyWalay, while `source` and `source_id` maintain the original platform identifiers. The `images` field uses JSONB format for flexible storage of property photos with metadata. Geographic coordinates (`latitude`, `longitude`) enable location-based features, while temporal fields (`created_at`, `updated_at`, `last_price_change_at`) support real-time synchronization and price tracking.

```
CREATE TABLE IF NOT EXISTS properties (
  our_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  source TEXT NOT NULL CHECK (source IN ('graana','lamudi','zameen')),
  source_id TEXT NOT NULL,
  source_human_id TEXT,
  title TEXT,
  prop_type TEXT,
  prop_subtype TEXT,
  area_size NUMERIC,
  area_unit TEXT,
  beds INT,
  baths INT,
```

```
    area_name TEXT,
    link TEXT,
    images JSONB,
    poc_name TEXT,
    poc_number TEXT,
    latitude NUMERIC,
    longitude NUMERIC,
    current_price NUMERIC,
    currency TEXT,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    last_price_change_at TIMESTAMPTZ
);

CREATE UNIQUE INDEX IF NOT EXISTS ux_properties_source_sourceid
    ON properties (source, source_id);

CREATE INDEX IF NOT EXISTS idx_properties_updated_at
    ON properties(updated_at DESC);
```

3.3.2 Property Price History Table

The `property_price_history` table implements comprehensive price tracking functionality, enabling users to monitor price changes over time and identify investment opportunities. Central to this table is the `our_id` foreign key, which links price records to specific properties, ensuring seamless integration and data consistency. The `changed_at` timestamp provides a chronological record of all price modifications, supporting the Investment Insights Module in generating trend analysis and market reports.

```
CREATE TABLE IF NOT EXISTS property_price_history (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    our_id UUID NOT NULL REFERENCES properties(our_id)
        ON DELETE CASCADE,
    price NUMERIC NOT NULL,
    currency TEXT,
    changed_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

CREATE INDEX IF NOT EXISTS idx_price_history_our_id_changed_at
    ON property_price_history(our_id, changed_at DESC);
```

3.3.3 Property Last Updates Table

The `property_last_updates` table serves as an efficient synchronization mechanism for tracking when properties were last modified on external platforms. This dedicated table enables the Integration Layer to quickly identify which properties require re-scraping without scanning the entire properties table, supporting the 1-hour maximum refresh rate requirement.

```
CREATE TABLE IF NOT EXISTS property_last_updates (
  our_id UUID PRIMARY KEY REFERENCES properties(our_id)
  ON DELETE CASCADE,
  last_updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

3.3.4 Saved Properties Table

The `saved_properties` table implements a many-to-many relationship between users and properties, enabling the property tracking and notification features. The composite primary key on `email` and `our_id` prevents duplicate saves while maintaining referential integrity. When a user saves a property, the Property Tracking Module automatically begins monitoring it for price changes and availability status, triggering notifications when updates occur.

```
CREATE TABLE IF NOT EXISTS saved_properties (
  email TEXT NOT NULL,
  our_id UUID NOT NULL REFERENCES properties(our_id)
  ON DELETE CASCADE,
  saved_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  PRIMARY KEY (email, our_id)
);
```

3.3.5 App Users Table

The `app_users` table maintains registered user profiles and serves as the authentication foundation for PropertyWalay. The `email` attribute, with its unique constraint, ensures single account per email address and serves as the primary identifier for user interactions across the system. This table is referenced by saved properties, search logs, reviews, and notifications, providing a cohesive framework for tracking user behavior that feeds into the AI Recommendation Engine.


```
CREATE TABLE IF NOT EXISTS app_users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email TEXT UNIQUE NOT NULL,  
  full_name TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

3.3.6 Property Reviews and Comments Tables

The `property_reviews` and `property_comments` tables facilitate community engagement and trust-building within the platform. The reviews table captures user ratings (1-5 scale enforced by CHECK constraint) and feedback, while the comments table enables discussions and inquiries. Both tables reference properties through `our_id` with CASCADE delete, ensuring referential integrity. These tables support transparency and reduce agent burden by allowing users to share experiences and ask questions publicly.

```
CREATE TABLE IF NOT EXISTS property_reviews (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  our_id UUID NOT NULL REFERENCES properties(our_id)  
    ON DELETE CASCADE,  
  reviewer_email TEXT NOT NULL,  
  rating INT CHECK (rating BETWEEN 1 AND 5),  
  comment TEXT,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX IF NOT EXISTS idx_property_reviews_our_id_created_at  
  ON property_reviews(our_id, created_at DESC);
```

```
CREATE TABLE IF NOT EXISTS property_comments (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  our_id UUID NOT NULL REFERENCES properties(our_id)  
    ON DELETE CASCADE,  
  commenter_email TEXT NOT NULL,  
  body TEXT NOT NULL,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX IF NOT EXISTS idx_property_comments_our_id_created_at  
  ON property_comments(our_id, created_at DESC);
```

3.3.7 Search Logs and Property Views Tables

The `search_logs` and `property_views` tables are structured to facilitate analytics and AI-powered recommendations within PropertyWalay. The `search_logs` table captures user queries and extracted filters in JSONB format, providing raw data for the NLU Module and trend analysis. The `property_views` table tracks engagement metrics, recording every property viewing event. Together, these tables bridge raw user interactions with actionable insights, underpinning the system's journey towards personalized recommendations and investment insights. The `filters` attribute in `search_logs` is particularly significant as it stores structured search parameters extracted by the NLU Module, enabling the AI Recommendation Engine to understand user preferences and refine suggestions.

```
CREATE TABLE IF NOT EXISTS search_logs (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT,
  query TEXT NOT NULL,
  filters JSONB,
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

```
CREATE INDEX IF NOT EXISTS idx_search_logs_created_at
ON search_logs(created_at DESC);
```

```
CREATE TABLE IF NOT EXISTS property_views (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  our_id UUID NOT NULL REFERENCES properties(our_id)
  ON DELETE CASCADE,
  viewer_email TEXT,
  viewed_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

```
CREATE INDEX IF NOT EXISTS idx_property_views_our_id_viewed_at
ON property_views(our_id, viewed_at DESC);
```

3.3.8 Price Alert Subscriptions Table

```
CREATE TABLE IF NOT EXISTS price_alert_subscriptions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  email TEXT NOT NULL,
  our_id UUID NOT NULL REFERENCES properties(our_id)
```

```
    ON DELETE CASCADE,  
    threshold_price NUMERIC NOT NULL,  
    currency TEXT,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    UNIQUE (email, our_id)  
);
```

3.3.9 Notifications Table

The notifications table is designed to streamline the handling of system-generated alerts and their delivery to users. Within this table, the `title` and `body` attributes provide human-readable content, while the `meta` field (JSONB format) stores structured data linking notifications to specific properties, actions, or events. The `is_read` boolean flag tracks notification status, enabling the user interface to distinguish between read and unread messages. Together, this schema provides a structured approach to connecting automated system events with user awareness, enhancing transparency and user engagement across the platform.

```
CREATE TABLE IF NOT EXISTS notifications (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    email TEXT NOT NULL,  
    title TEXT NOT NULL,  
    body TEXT,  
    meta JSONB,  
    is_read BOOLEAN NOT NULL DEFAULT FALSE,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()  
);
```

```
CREATE INDEX IF NOT EXISTS idx_notifications_email_created_at  
    ON notifications(email, created_at DESC);
```

3.3.10 Relational Integrity

The PropertyWalay database schema employs foreign key constraints with appropriate cascading behaviors to maintain referential integrity. The `saved_properties` table establishes a relationship with `app_users` through an added constraint that uses `ON DELETE SET NULL`, preserving analytics data while respecting user deletion requests.

```
ALTER TABLE IF EXISTS saved_properties
```

```
ADD CONSTRAINT IF NOT EXISTS fk_saved_properties_email  
FOREIGN KEY (email) REFERENCES app_users(email)  
ON DELETE SET NULL;
```

The schema supports all functional requirements including real-time listings integration, property tracking and notifications, AI-powered recommendations, investment insights, and user history management. The use of PostgreSQL's native features such as JSONB for semi-structured data and TIMESTAMPTZ for timezone-aware timestamps ensures the system can handle the diverse requirements of Pakistan's real estate market across multiple cities and time zones.